

BACKTRACK: pitch and Q&A

One sentence

BACKTRACK helps a learner find a missing step, practice it with Khan Academy, and return to the lesson they need now.

Ten seconds

Stuck on today's lesson, but unsure what to review? BACKTRACK finds an earlier step, connects it to Khan practice, and checks your way back.

Thirty seconds

A learner gets stuck on today's equation and hears "review." But review what? BACKTRACK checks the steps underneath that goal. If the learner demonstrates one, its review disappears. If they need help, the route connects them to a specific Khan lesson and exercise. Fresh problems check the return. Our proposed school pilot will measure who gets back to the lesson and can still do it 7-14 days later.

Five-minute pitch script

Use this as a rehearsal script, not words to read from slides. The deck has 15 pages including sources. Rehearse to finish in about 4:45, leaving room for pauses. Keep the live interaction brief and have the static route diagrams ready if connectivity fails.

"Picture a learner with one hard homework question and a short study break. They can search the topic and watch a lesson, yet still not know which step is missing. We built BACKTRACK to make that next decision easier.

A GPS keeps your destination while recalculating the route. BACKTRACK starts with the task the learner needs now and looks for a useful place to work.

Now imagine a learner looking at this quadratic equation. Their class is already here. Something underneath it stopped making sense. The advice is to review. But review what?

The difficulty could be finding factors. It could be multiplying brackets. It could be knowing that each factor can equal zero. Giving every learner the same review ignores those differences.

BACKTRACK starts with the destination. A learner or teacher chooses the goal they need now. Their answer can give a wrong-turn clue. For the same quadratic, one learner may have the right factor numbers but the wrong signs, while another has the wrong pair. Those answers lead to different starting checks. The route explains the choice and tests it.

That is our first product moment: the reward is less unnecessary review. We don't invent minutes saved. Two fresh answers can remove an actual review stop. A learner who needs a repair gets a different reward: they can demonstrate something they couldn't demonstrate before.

The learning stop is small and specific: one idea, a worked example, then a chance to try it. Official Khan videos play the relevant segment, with an option to continue. Original Khan exercises support further practice. Fresh BACKTRACK checks bring the learner back to the goal.

The next slide invites you to try the demo. Click the button or scan the code. Try each of the two wrong answers and see why they get different starting checks.

The destination then comes back into focus. The learner tries a fresh current-level task. That gives us a clear question to test in school: did this route help them return, and did that return last?

Our signature measure is retained reentry. Among learners initially blocked at the selected destination, how many pass a fresh post-route assessment and still pass a parallel task seven to fourteen days later? The denominator includes every eligible learner enrolled in the evaluation. Missing follow-up will be visible.

From November through March, we propose eighty to one hundred twenty learners across two or three cohorts, subject to school agreements, permissions, and access. Teachers will confirm the maps and assessments. A recurring lab, remediation, or homework window will fit the school's routine.

We want a fair comparison: the BACKTRACK workflow against teacher-selected Khan practice, with similar time and support. Where feasible, a randomized delayed start lets both groups receive the route while preserving a comparison window. If that design isn't possible, we'll report feasibility rather than pretend a pre-and-post change proves causation.

The route is also the reward. Knowing a step means its review can leave the route. Learning a new step means a visible gap closes. Coming back later means continuing with earlier work saved. That is a practical reason to keep learning: progress changes what you need to do next.

The replication unit is a destination kit: the goal, reviewed map, checks, Khan material, return assessments, and facilitator guide. We will ask a second educator to use it and measure how much help they need.

A one-hundred-thousand-peso implementation scenario prioritizes learning review, access, school deployment, and evaluation. Learner access stays free. We will measure costs and interview actual budget owners before inventing a school price.

We are the University of the Philippines Manila BACKTRACK team. Our next step is to run this recovery routine with schools and measure the return.

By March, we want to show more than another account created. We want a learner who found a route back to today's lesson, and evidence that the learning stayed."

Demonstration rehearsal

Open the deployed /demo route before the slot. Tap the sample 3 and 4: the route starts with signs and solutions. Replay and try 2 and 6: the route starts with factor pairs. Continue and open the matching short Khan repair. Do not manufacture a real learner story. Use the static deck to show later states if time is short. The app's working answer-driven path is distinct from pilot evidence.

Hostile questions and answers

Isn't this just adaptive learning? Yes, it belongs to that family. The proposed contribution is a specific Khan-centered recovery routine, including a current-class return outcome. We must demonstrate value beyond a simpler routine.

Doesn't Khan already personalize? It does. Learning Paths and mastery tools are real. We start from a selected current destination and test the need for this lightweight entry point. We should not deploy BACKTRACK where an existing Khan workflow serves the school better.

Why not Gemini? Gemini can diagnose and plan study. Our difference is a reviewed, inspectable route and school implementation around Khan practice, with a formal reentry test. That is a hypothesis about useful delivery, not a claim that Gemini cannot imitate a feature.

Why Khan rather than any video site? The proposed school routine uses coherent explanations, exercises, assignments, and teacher reports in one existing learning environment. A video player alone would not satisfy our integration claim.

Where does Khan data come from? The public experience records resource opens and learner reports locally. It does not receive Khan results. In the pilot, authorized teachers will use available assignment reports/exports and reconcile them with pseudonymous records. Availability must be verified before enrollment.

How do you know the gap caused the difficulty? We do not establish causation from a wrong answer. It suggests a prerequisite to investigate. Targeted checks, later performance, teacher review, and a controlled comparison strengthen the evidence. The wording must remain "suspected blocker" until justified.

What does shortest mean? A small useful evidence-supported route to the chosen destination. We do not solve a globally optimal shortest-path problem or guarantee minimum time.

What if confidence is wrong? Correctness is decided by mathematics. Confidence changes follow-up or support. It never grants mastery or changes the score. We will evaluate whether the extra question is useful enough to keep.

What if they use AI to cheat? Remote practice cannot reliably establish who produced an answer. The interface explains that misleading answers produce an unhelpful route. Formal outcome tasks are separately supervised. We do not claim cheating detection.

Does two correct answers prove mastery? No. It supports a limited route decision. Formal reentry uses reviewed fresh tasks and a delayed check. Broader and longer-term mastery requires more evidence.

Who builds the graph? The team authors destination kits and educators review dependencies, discriminating questions, alternate methods, and curriculum fit. Versioning records changes. Content review is a real scaling cost.

How does it know today's lesson? A teacher or learner selects the destination. The public tool does not read an LMS or infer a class schedule automatically.

Why another tool for teachers? It must earn its place. A teacher shares a destination and uses an existing Khan class. The pilot measures the new setup/reporting burden and compares with teacher-selected practice. If it costs more effort without useful gains, simplify or stop that workflow.

What about no internet? Local progress can persist and the open page can continue. Khan content and initial loading still need a connection. Use scheduled access or local worked examples and record excluded/delayed learners. There is no full offline Khan claim.

What if the gap takes hours? The route can pause and resume. Short sessions do not erase deep prerequisites. Repeated difficulty leads to human support, not a guaranteed quick fix.

How will you establish impact? With baseline eligibility, fresh post tasks, delayed tasks, a clear denominator, and a feasible comparison. Public engagement and unsupervised checks are not added to formal school outcomes.

Who pays after March? That is unvalidated. We will measure actual costs and interview institutional budget owners in university extension, schools, nonprofits, LGUs, and CSR. We will not claim an invented package price proves sustainability.

How do you protect minors? School approval, required parent/guardian permissions, adult oversight, minimal data, school-controlled identity keys, restricted records, aggregate reporting, and an institutional review process. We do not claim blanket legal compliance from the software design.

Why does this deserve attention? A learner can have access to excellent lessons and still be unable to choose useful help. BACKTRACK makes that decision visible and testable: the mistake informs a check, the repair fits the step, and the learner returns to the task. The opportunity is to make existing Khan resources easier to use at the moment of difficulty.

Weak answers to avoid

"Nobody has done this." "Our AI finds the exact gap." "A video open proves meaningful use." "Two correct answers establish mastery." "We will reach thousands." "Schools will pay PHP 25,000." "Our design makes learning addictive." "We already partner with Khan." None of these is established.